

St Aloysius' College
Year 12 Mid-Year Examinations
2008

EXTENSION 2 MATHEMATICS
(Additional paper)

Total marks - 80

Reading time - 5 minutes

Working time – 2 hours

Examination papers must NOT be removed
from the examination room.

General Instructions

- Start each question in a new booklet
- Board approved calculators may be used
- Marks may be deducted for careless or badly arranged work
- Show all necessary work

SUPERVISOR'S INSTRUCTIONS:

Please issue five 4 page answer booklets.

Please collect the examination paper with the answer booklets.

Question 1 (16 marks)

(a) Let $\mathcal{H}$ be the hyperbola $\frac{x^2}{5^2} - \frac{y^2}{3^2} = 1$

(i) Calculate the eccentricity of $\mathcal{H}$ 1

(ii) Find the coordinates of the foci and the directrices of $\mathcal{H}$ 3

(iii) Find the equations of the asymptotes of $\mathcal{H}$ 1

(b) The point $P(x_1, y_1)$ lies on the ellipse $\frac{x^2}{9} + \frac{y^2}{7} = 1$.

The tangent at P cuts the x axis at T , and has equation $\frac{x_1 x}{9} + \frac{y_1 y}{7} = 1$

(i) Find the coordinates of T . 1

(ii) Using the focus-directrix definition, or otherwise, show that

$$\frac{PS}{PS'} = \frac{TS}{TS'} \quad \text{where } S \text{ and } S' \text{ are the foci of the ellipse.}$$
 3

(c) (i) Write down the equation of the chord of contact from $T(x_0, y_0)$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$. 1

(ii) Can the chord of contact ever be a diameter? 1

(d) (i) Let x be a fixed number satisfying $0 < x < 1$. Use the method of mathematical induction to prove that 3

$$(1-x)^n > 1-nx \quad \text{for } n=2,3,\dots$$

(ii) Deduce that 2

$$\left(1 - \frac{1}{3n}\right)^n > \frac{2}{3} \quad \text{for } n=2,3,\dots$$

Question 2 (16 marks) START A NEW BOOKLET

(a) (i) Express $-1 - i\sqrt{3}$ in modulus-argument form. 2

(ii) Hence evaluate $(-1 - i\sqrt{3})^5$ in the form $a + ib$ 2

(b) Sketch the region where the inequalities 3

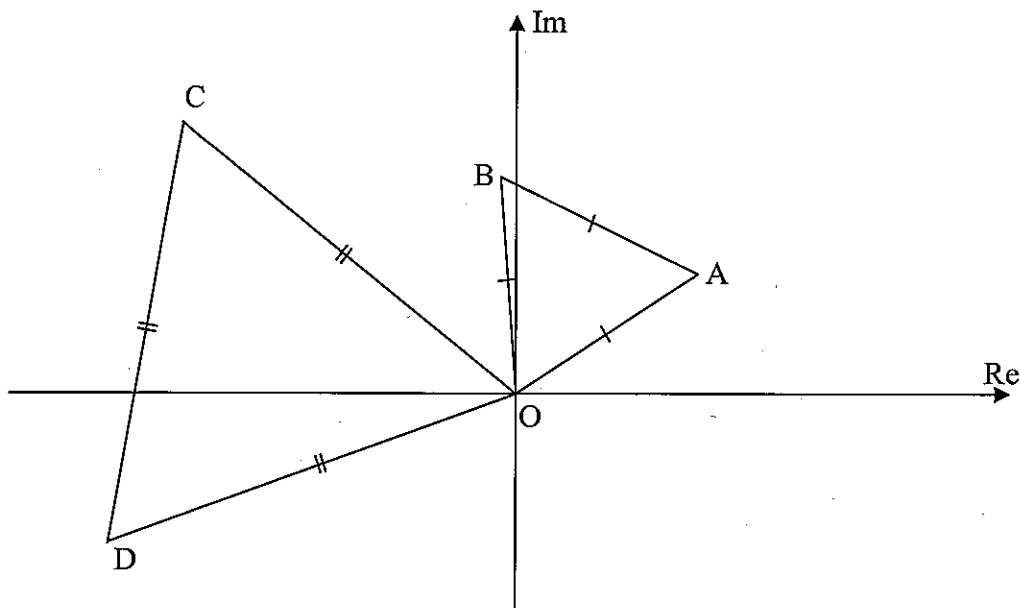
$$|z - 4 - i| \leq 5 \quad \text{and} \quad |z + 2| \leq |z - 2| \quad \text{both hold.}$$

(c) Let $z_1 = \frac{4-3i}{5}$ and $z_2 = \frac{8+15i}{17}$ so that $|z_1| = |z_2| = 1$

(i) Find $z_1 z_2$ and $\bar{z}_1 z_2$ in the form $a + ib$ 2

(ii) Hence find two distinct ways of writing 85^2 as the sum $x^2 + y^2$, 2
where x and y are positive integers.

(d)



Let $\omega = cis \frac{\pi}{3}$ and the points A, B, C and D correspond to the complex numbers α, β, γ and δ respectively. $\triangle OAB$ and $\triangle OCD$ are equilateral.

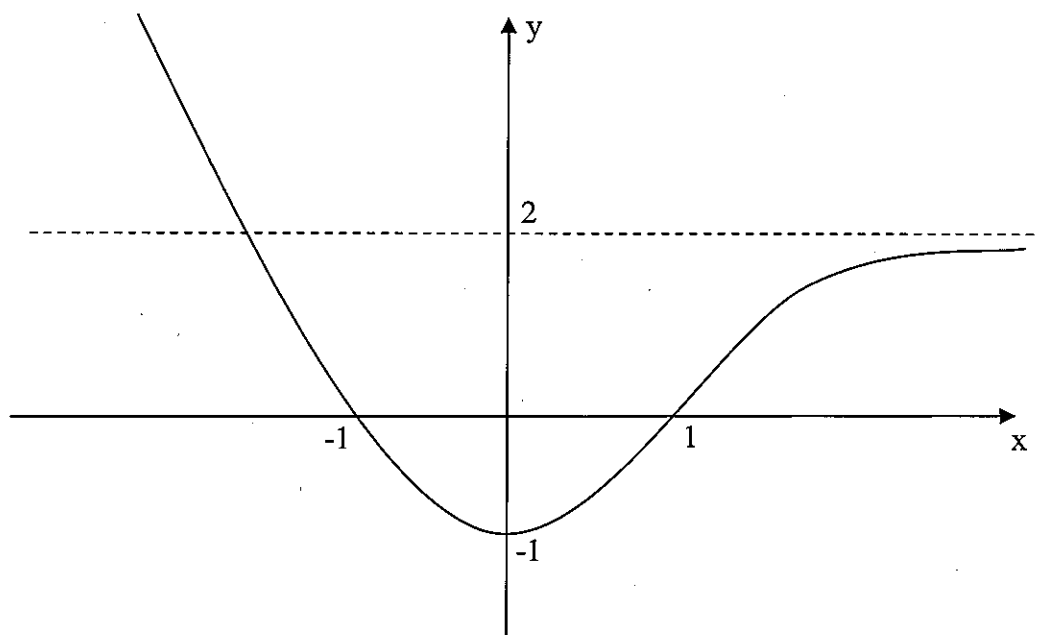
(i) Explain why $\delta = \omega\gamma$ 2

(ii) Find the complex number α in terms of β 1

(iii) Using complex numbers, show that the lengths AC and BD are equal. 2

Question 3 (16 marks) START A NEW BOOKLET

The function $y = f(x)$ is defined by the following graph.



On separate diagrams, provide sketches of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = \frac{1}{f(x)}$ | 3 |
| (ii) | $y = f(x)$ | 2 |
| (iii) | $y = f(x) $ | 2 |
| (iv) | $y^2 = f(x)$ | 3 |
| (v) | $y = \ln f(x)$ | 3 |
| (vi) | $y = f(\ln x)$ | 3 |

Question 4 (16 marks) START A NEW BOOKLET

- (a) (i) Derive the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$. 2
- (ii) Derive the equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point $P(a \sec \theta, b \tan \theta)$. 2
- (iii) Show that the point of intersection of these tangents is $(a, b \tan \frac{\theta}{2})$. 3
- (b) Find in mod-arg form the values of z which satisfy $z^5 = \frac{1+i}{\sqrt{2}}$ 4
- (c) The sequence of numbers 1, 3, 7, 17, ... is defined by 5
 $T_1 = 1, T_2 = 3, T_{n+2} = 2T_{n+1} + T_n$ for $n \geq 1$
Use mathematical induction to prove that $T_n = \frac{1}{2}(1 + \sqrt{2})^n + \frac{1}{2}(1 - \sqrt{2})^n$ for $n \geq 1$

Question 5 (16 marks) START A NEW BOOKLET

- (a) Find real numbers a, b and c such that $\frac{x^2 - x}{2(x+1)} = ax + b + \frac{c}{x+1}$ 3
- (b) Consider the function $y = \frac{x^2 - x}{2(x+1)}$
- (i) Using part (a), or otherwise, write down the equations of the oblique asymptote and the vertical asymptote. 2
- (ii) Find the intersection of the asymptotes, 1
- (iii) Find the coordinates and the nature of any stationary points. 5
- (iv) Sketch the function taking care with the relative positions of the critical features. 4
- (v) For what two values of k does the equation $kx = \frac{x^2 - x}{2(x+1)}$ have one distinct solution only. 1

Q2

a) $\frac{x^2}{5^2} - \frac{y^2}{3^2} = 1$

(i) $b^2 = a^2(e^2 - 1)$

$\therefore 9 = 25(e^2 - 1)$

$\therefore e^2 = \frac{9}{25} + 1$

$e = \frac{\sqrt{34}}{5}$ (1)

(ii) $ae = \sqrt{34}$

$\therefore S(\sqrt{34}, 0) S(-\sqrt{34}, 0)$ (1)

and $\frac{a}{e} = \frac{25}{\sqrt{34}}$

$\therefore x = \pm \frac{25}{\sqrt{34}}$ (1)

and (1) for twins.

(iii) $\frac{y^2}{9} = \frac{x^2}{25} - 1$

$y = \pm \frac{3}{5} \sqrt{x^2 - 25}$

$\therefore$ Asymptotes $y = \pm \frac{3x}{5}$ (1)

b) (i) $\frac{x_1 x_2}{9} + \frac{y_1 y_2}{7} = 1$ Let $y = 0 \Rightarrow x = \frac{9}{x_1} \therefore T\left(\frac{9}{x_1}, 0\right)$ (1)

(ii) $\frac{PS}{PS'} = \frac{\angle PN}{\angle PN'}$ where N, N' are feet of perpendicular from P to the directrices. (1)

$= \frac{\frac{3}{e} - x_1}{\frac{3}{e} + x_1}$

$= \frac{3 - ex_1}{3 + ex_1}$ (1)

But $\frac{TS}{TS'} = \frac{\frac{9}{x_1} - 3e}{\frac{9}{x_1} + 3e}$

$= \frac{3 - ex_1}{3 + ex_1}$ (1)

$\therefore \frac{PS}{PS'} = \frac{TS}{TS'}$

c) (i) $\frac{xx_0}{16} + \frac{yy_0}{4} = 1$ (1)

(ii) No because $(0, 0)$ does not satisfy the above equation. (1)

d) (i) Let $S(n)$ be the statement that $(1-x)^n > 1-nx$ for $n=2, 3, \dots$ and $0 < x < 1$
For $n=2$ LHS $= 1 - 2x + x^2 > 1 - 2x$ $\therefore S(2)$ is true. (1)

Assume $S(k)$ i.e. $(1-x)^k > 1-kx$ for $k=2, 3, \dots$

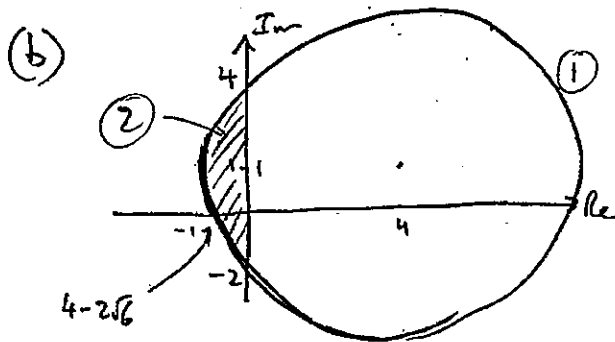
Consider $S(k+1)$ LHS $= (1-x)^{k+1}$
 $= (1-x)^k (1-x)$
 $> (1-kx)(1-x)$ by assumption (1)
 $= 1 - (k+1)x + kx^2$
 $> 1 - (k+1)x$ (1)
 $=$ RHS

Hence $S(n)$ is true for all integers $n \geq 2$ by the Principle of M.I.

(ii) Let $x = \frac{1}{3n}$ (1)
 $\therefore 0 < x < 1$ for $n=2, 3, \dots$
 $\therefore$ Part (i) $\Rightarrow \left(1 - \frac{1}{3n}\right)^n > 1 - \frac{1}{3}$ (1)
 $= \frac{2}{3}$

Q2 (a) (i) $-1 - i\sqrt{3} = 2 \operatorname{cis}\left(-\frac{2\pi}{3}\right)$ (2)

(ii) $(-1 - i\sqrt{3})^5 = 2^5 \left[\cos \frac{-10\pi}{3} + i \sin \frac{-10\pi}{3} \right]$ [De Moivre] (1)
 $= -16 + 16\sqrt{3}i$ (1)



(c) (i) $z_1 z_2 = \frac{4-3i}{5} \cdot \frac{8+15i}{17}$
 $= \frac{77+36i}{85}$ (1)

$\bar{z}_1 \bar{z}_2 = \frac{4+3i}{5} \cdot \frac{8+15i}{17}$
 $= \frac{-13+84i}{85}$ (1)

(ii) $|z_1| = |z_2| = 1 \therefore |z_1 z_2| = 1$ and $|\bar{z}_1 \bar{z}_2| = 1$

$\therefore 77^2 + 36^2 = 85^2 \Rightarrow x = 77, y = 36$ (1)

and $13^2 + 84^2 = 85^2 \Rightarrow x = 13, y = 84$ (1)

(d) (i) $|w| = 1 \therefore |s| = |w\gamma|$ since $OD = OC$ (1)
 and $\angle COD = \frac{\pi}{3} \therefore \arg s = \arg \gamma + \frac{\pi}{3}$ [Not principle]
 $= \arg \gamma + \arg w$
 $= \arg w\gamma$ (1)

$\therefore s = w\gamma$

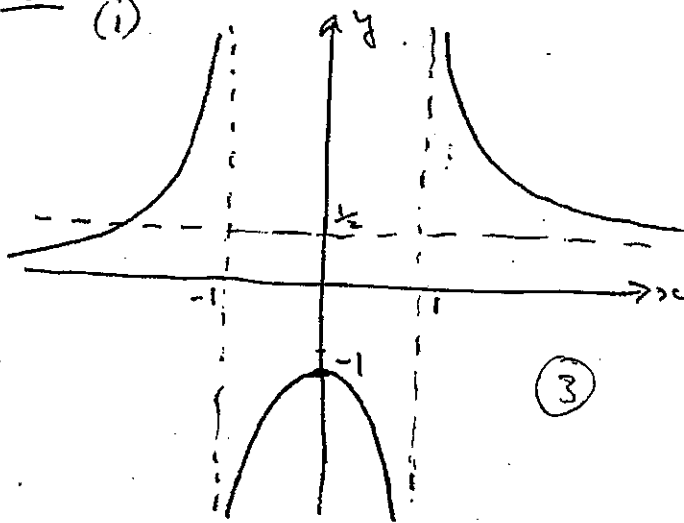
(ii) $\beta = w\alpha \Rightarrow \alpha = \frac{\beta}{w}$
 $= \bar{w}\beta$ (1)

(iii) $AC = |\gamma - \alpha|$ and $BD = |s - \beta|$
 $= |\gamma - \bar{w}\beta|$
 $= |\bar{w}| |w\gamma - \beta|$ (1)
 $= |w\gamma - \beta|$ (1)

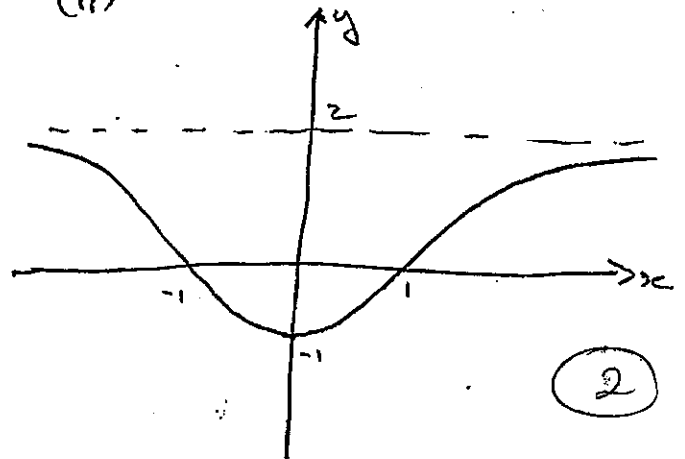
$\therefore AC = BD$

Q3

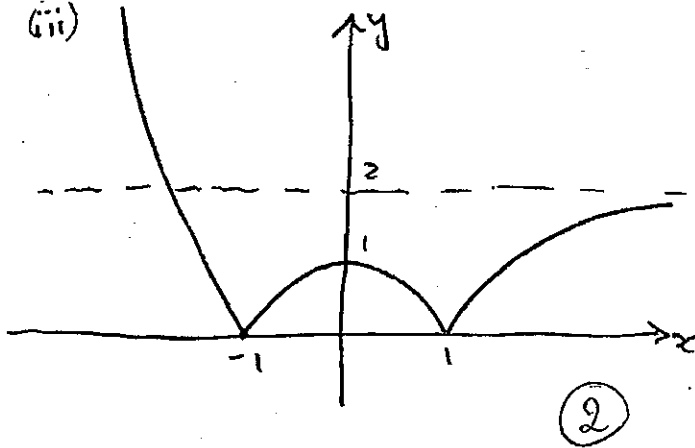
(i)



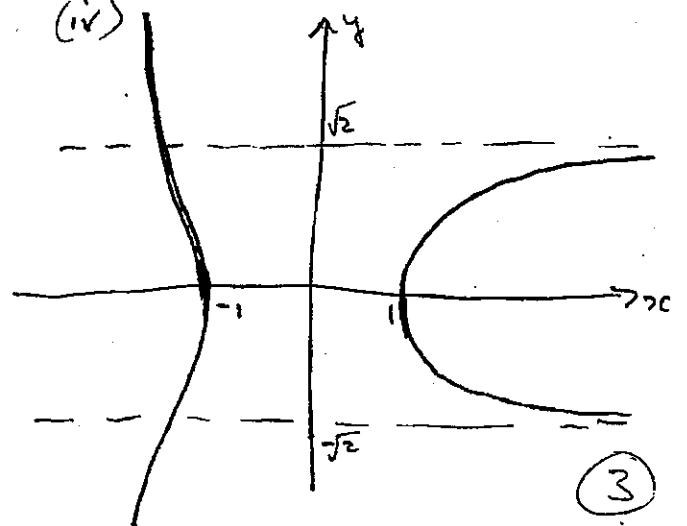
(ii)



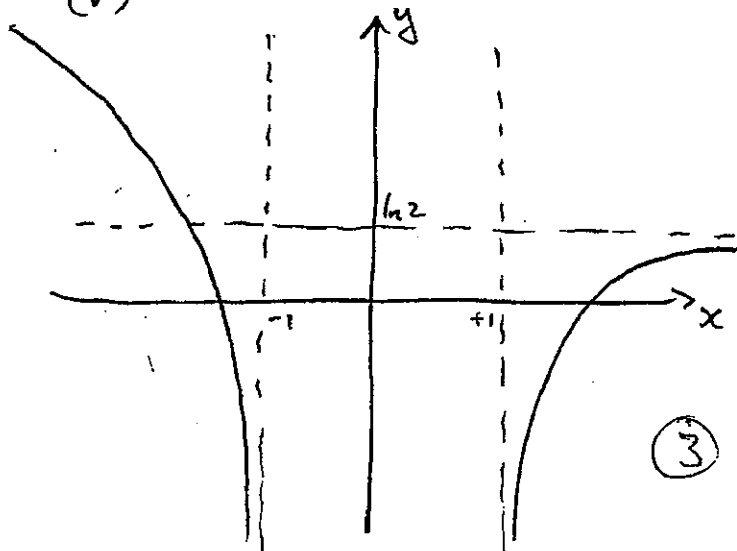
(iii)



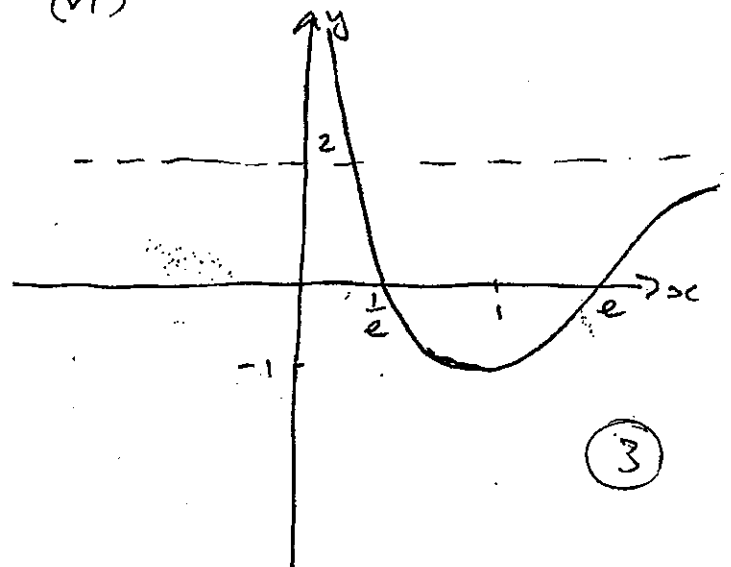
(iv)



(v)



(vi)



Q4 (a) (i) $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$
 $= \frac{-b \cos \theta}{a \sin \theta}$ (1)

$\therefore y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$

$\therefore \frac{y \sin \theta}{b} - \sin^2 \theta = -\frac{x \cos \theta}{a} + \cos^2 \theta$

$\therefore \frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$ (A) (1)

(iii) (A) + (B) $\times \cos \theta \Rightarrow \frac{x \cos \theta}{a} + \frac{y \cos \theta}{a} = 1 + \cos \theta$

$x \left(\frac{\cos \theta + 1}{a} \right) = 1 + \cos \theta$

$\therefore x = a$ (1)

$\rightarrow (A) \Rightarrow \frac{y \sin \theta}{b} = 1 - \cos \theta$

$y = \frac{b(1 - \cos \theta)}{\sin \theta}$ (1)

$= \frac{b \times 2 \sin^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}$

$= b \tan \frac{\theta}{2}$ (1)

b) Since $\left| \frac{1+i}{\sqrt{2}} \right| = 1 \therefore$ Let $z = \cos \theta$ (1)

$\therefore \cos 5\theta + i \sin 5\theta = \frac{1+i}{\sqrt{2}}$ using De Moivre

$\therefore \cos 5\theta = \frac{1}{\sqrt{2}}$ and $\sin 5\theta = \frac{1}{\sqrt{2}}$ (1)

$\therefore 5\theta = \frac{\pi}{4}, -\frac{7\pi}{4}, \frac{9\pi}{4}, -\frac{15\pi}{4}, \frac{17\pi}{4}$

$\theta = \frac{\pi}{20}, -\frac{7\pi}{20}, \frac{9\pi}{20}, -\frac{3\pi}{4}, \frac{17\pi}{20}$ (1)

$z = \cos \frac{\pi}{20}, \cos -\frac{7\pi}{20}, \cos \frac{9\pi}{20}, \cos -\frac{3\pi}{4}, \cos \frac{17\pi}{20}$ (1)

(or $45^\circ, -315^\circ, 405^\circ, -675^\circ, 765^\circ$)

(or $2.25^\circ, -15.75^\circ, 20.25^\circ, -33.75^\circ, 38.25^\circ$)
 (or $9^\circ, -63^\circ, 81^\circ, -135^\circ, 153^\circ$)

c) See over.

(i) or $\frac{2x}{a^2} + \frac{2y \cdot y'}{b^2} = 0$
 $y' = -\frac{b^2 x}{a^2 y}$

(ii) or $\frac{2x}{a^2} - \frac{2y \cdot y'}{b^2} = 0$
 $y' = \frac{b^2 x}{a^2 y}$

Q4 c) For $T_1 = 1$; $T_2 = 3$; $T_{n+2} = 2T_{n+1} + T_n$ $n \geq 1$

RTP $T_n = \frac{1}{2}(1+\sqrt{2})^n + \frac{1}{2}(1-\sqrt{2})^n$ $n \geq 1$

For $n=1$ $T_1 = \frac{1}{2}(1+\sqrt{2}) + \frac{1}{2}(1-\sqrt{2})$
 $= 1$

For $n=2$ $T_2 = \frac{1}{2}(3+2\sqrt{2}) + \frac{1}{2}(3-2\sqrt{2})$
 $= 3$

$\therefore$ True for $n=1$ and $n=2$

Assume that

$$T_k = \frac{1}{2}(1+\sqrt{2})^k + \frac{1}{2}(1-\sqrt{2})^k$$

and $T_{k+1} = \frac{1}{2}(1+\sqrt{2})^{k+1} + \frac{1}{2}(1-\sqrt{2})^{k+1}$

for any integer $k \geq 1$

Now $T_{k+2} = 2T_{k+1} + T_k$

$$= (1+\sqrt{2})^{k+1} + (1-\sqrt{2})^{k+1} + \frac{1}{2}(1+\sqrt{2})^k + \frac{1}{2}(1-\sqrt{2})^k \quad \text{by assumption}$$

$$= \frac{1}{2}(1+\sqrt{2})^k (2+2\sqrt{2}+1) + \frac{1}{2}(1-\sqrt{2})^k (2-2\sqrt{2}+1)$$

$$= \frac{1}{2}(1+\sqrt{2})^k (1+\sqrt{2})^2 + \frac{1}{2}(1-\sqrt{2})^k (1-\sqrt{2})^2$$

$$= \frac{1}{2}(1+\sqrt{2})^{k+2} + \frac{1}{2}(1-\sqrt{2})^{k+2} \quad \text{as required.}$$

Hence if the proposition is true for any two ~~successive~~ consecutive integers then it is true for the next integer.

Since it is true for $n=1$ and $n=2$, therefore it is true for all $n \geq 1$ by the Principle of Mathematical Induction.

Q5 a) $axc + b + \frac{c}{x+1} = \frac{axc^2 + (a+b)x + b+c}{x+1}$

(3)

$\therefore$ For the required equality $a = \frac{1}{2}$, $b = -1$, $c = 1$

b) (i) $y = \frac{x}{2} - 1$ and $x = -1$

(2)

(ii) $(-1, -\frac{3}{2})$

(1)

(iii) $y' = \frac{2(2x-1)(x+1) - 2(x^2-x)}{2(x+1)^2}$
 $= \frac{x^2 + 2x - 1}{2(x+1)^2}$

(1)

(2)

$y' = 0 \Rightarrow x = -1 \pm \sqrt{2}$

For $x = \sqrt{2} - 1$, $y = \sqrt{2} - \frac{3}{2}$

(1)

x	0	$\sqrt{2}-1$	1
y'	$-\frac{1}{2}$	0	$+\frac{1}{4}$

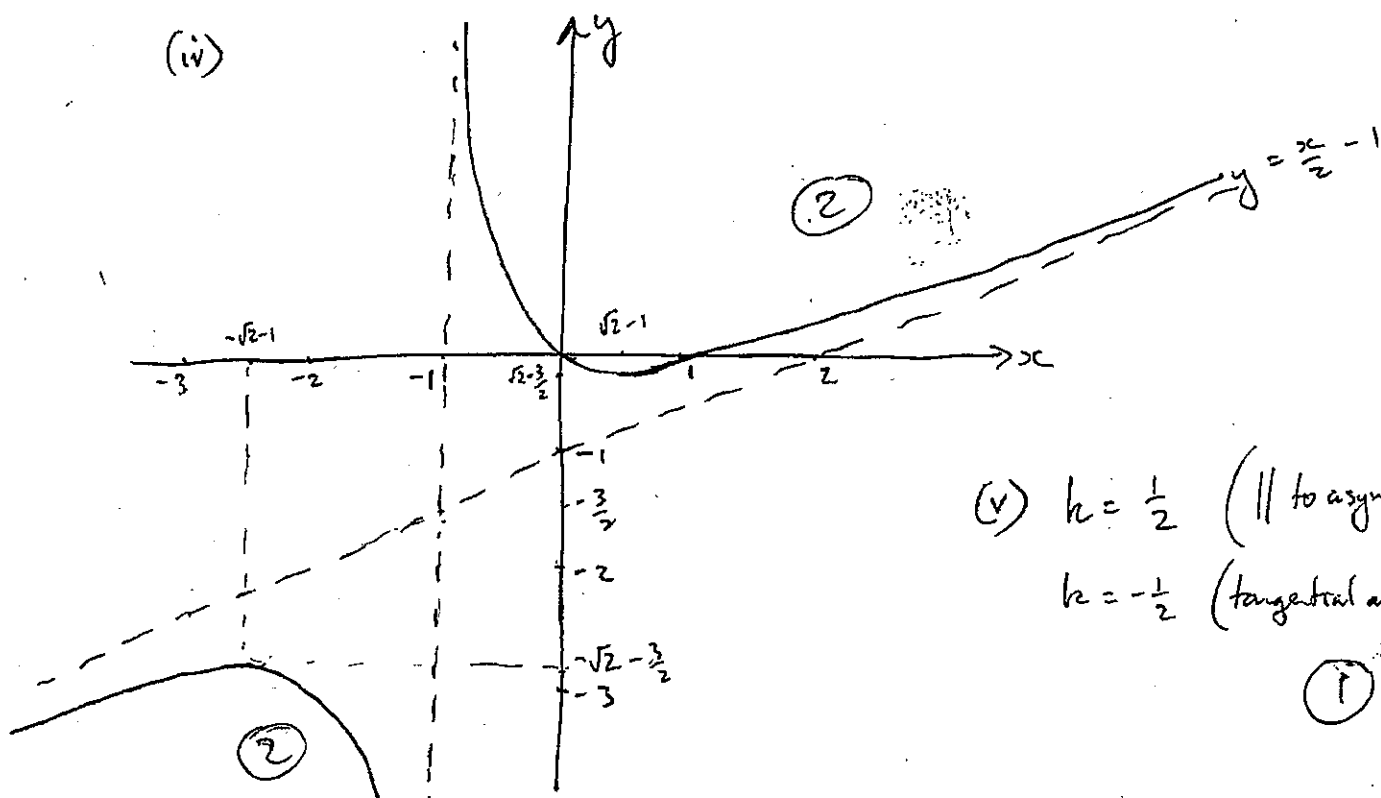
Minimum turning point. (1)

For $x = -\sqrt{2} - 1$, $y = -\sqrt{2} - \frac{3}{2}$

x	-3	$-\sqrt{2}-1$	-2
y'	$+\frac{1}{4}$	0	$-\frac{1}{2}$

Maximum turning point.

(iv)



(v) $h = \frac{1}{2}$ (|| to asymptote)
 $h = -\frac{1}{2}$ (tangent at 0)

(1)